

Volumes in Calabi-Yau Complete Intersection of Products of Projective Space

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Abstract. We prove that the birational automorphism group of a general Calabi-Yau complete intersection X given by ample divisors in $\mathbb{P}^{n_1} \times \cdots \times \mathbb{P}^{n_l}$ is always Lorentzian. Applying the Kawamata-Morrison cone theorem on such X , we compute $\text{vol}_X(D + sA)$ for any divisor $D \in \partial\overline{\text{Eff}}(X)$ and ample divisor A when s is small. We also provide examples of volumes of certain Cartier divisors that involve the digamma function.

1 Introduction

It is conjectured in [Mor93] and [Kaw97] that the movable effective cone of a Calabi-Yau manifold has a rational fundamental domain under the action of the birational automorphism group.

Conjecture 1.0.1 (*Kawamata-Morrison cone conjecture*) *Let X be a Calabi-Yau manifold. Then there exists a rational polyhedral fundamental domain Π for the action of the birational automorphism group $\text{Bir}(X)$ on the movable effective cone $\overline{\text{Mov}}^e(X)$ in the sense that*

- (i) $\overline{\text{Mov}}^e(X) = \bigcup_{g \in \text{Bir}(X)} g^*\Pi$,
- (ii) $\text{int}(\Pi) \cap \text{int}(g^*\Pi) = \emptyset$ if $g^* \neq \text{id}$.

The conjecture has been proven in the case when X is a general Wehler N -fold i.e. a general hypersurface of multidegree $(2, \dots, 2)$ in $(\mathbb{P}^1)^{N+1}$ for $N \geq 3$ ([CO15, Theorem 1.3]). In [FLT23], the structure of the boundary of the pseudoeffective cone $\overline{\text{Eff}}(X)$ has been studied. The divergent-recurrent decomposition of $\partial\overline{\text{Eff}}(X)$ is proven in [FLT23, Theorem 2.4.2], and the following result on the asymptotic behavior of volume function near $\partial\overline{\text{Eff}}(X)$ is derived:

Theorem 1.0.2 ([FLT23, Theorem 1.3.7]) *Suppose X is a general Wehler Calabi-Yau N -fold. Then, for every pseudoeffective $\mathbb{R}$ -divisor D in $\partial\overline{\text{Eff}}(X)$ and sufficiently ample divisor A , there exists an integer $s(D) \in \{0, \dots, N-2\}$ and a real number $\delta(D) \in [1, \frac{1}{2}(N - s(D))]$ such that*

$$\liminf_{s \downarrow 0} \frac{\log \text{vol}(D + sA)}{\log s} = \delta(D),$$

$$\limsup_{s \downarrow 0} \frac{\log \text{vol}(D + sA)}{\log s} = \frac{N - s(D)}{2}.$$

The fact that the birational automorphism group of a general Wehler N -fold is Lorentzian provides a hyperbolic subspace in $\mathbb{P}N^1(X)$, and plays an important role when investigating the structure of $\partial\overline{\text{Eff}}(X)$ (see [FLT23, §2.3]). Let $\mathbf{n} = (n_1, \dots, n_l) \in \mathbb{N}$ such that $|\mathbf{n}| \geq 4$ and $\mathbf{n} \neq (2, 2)$. In [Yá22], the author generalized the results in [CO15], and proved the Kawamata-Morrison cone conjecture when X is a general Calabi-Yau complete intersection in $\mathbb{P}^{\mathbf{n}} := \mathbb{P}^{n_1} \times \cdots \times \mathbb{P}^{n_l}$ given by the intersection of n ample divisors, with $n \leq \min\{n_i\}$. He also conjectured that the